

Week 04 · Transformers without heavy mathematics

OPEN / WEEK 04

The confusing sentence

In “Ava told Bea that she won,” which earlier word does “she” refer to? A model must use context, yet ambiguity can remain.

Which earlier token can each word use?

	I	read	books
I	1.00	0.00	0.00
read	0.30	0.70	0.00
books	0.25	0.65	0.10

The mask hides future positions during generation

First vote: what would you do?

QUESTION / WEEK 04

In “Ava told Bea that she won,” which earlier word does “she” refer to? A model must use context, yet ambiguity can remain.

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

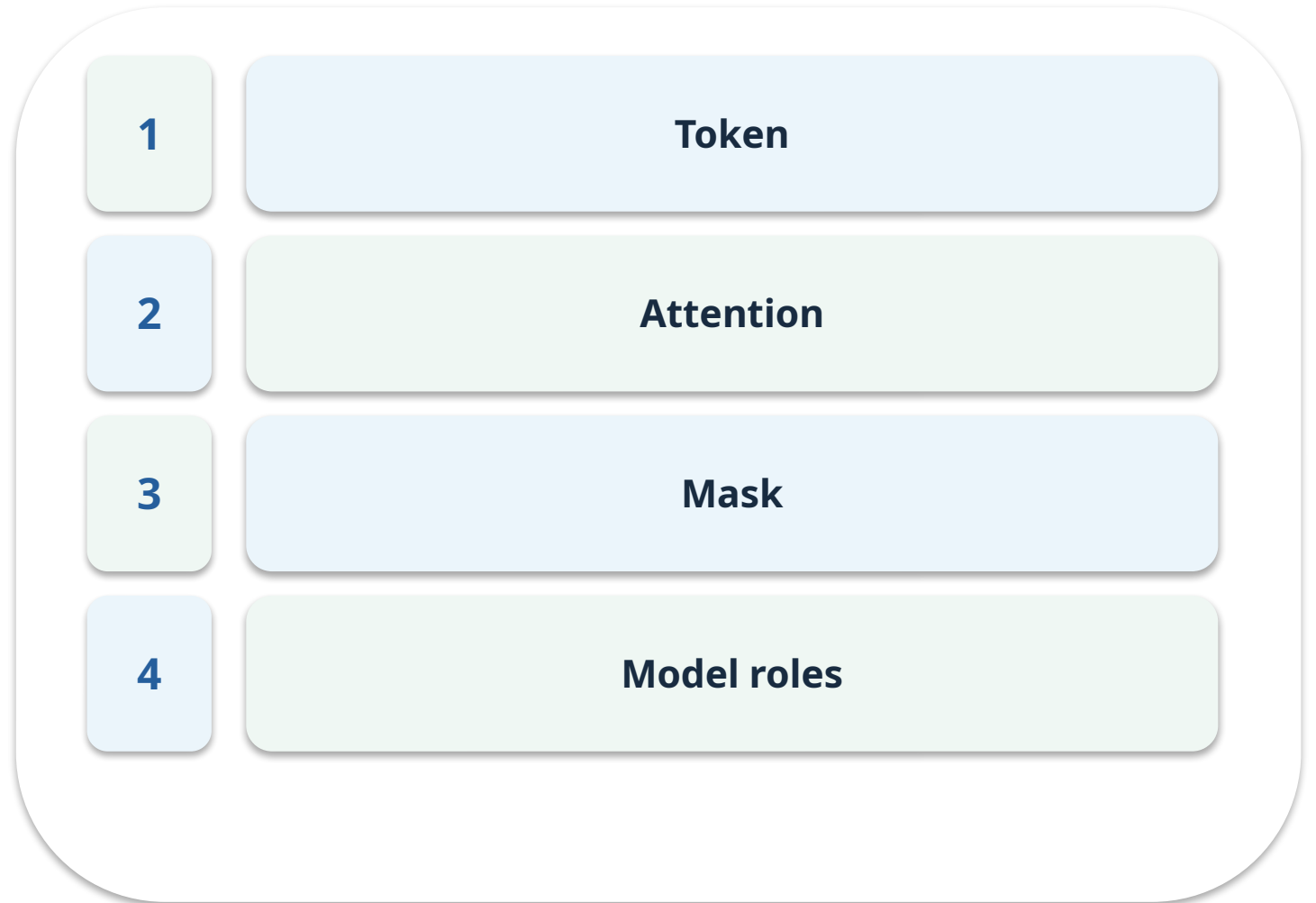
B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 04

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.



1. Predict · Token

QUESTION / WEEK 04

Why can word count differ from token count?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A A tokenizer may split a word into several pieces.

B One word always equals one token.

Which source or test could settle this?

Token • a familiar picture

ANALOGY / WEEK 04

A sentence is cut into tiles before a game reads it.

What matches our case, and where does the analogy stop?

A sentence is cut into tiles before a game reads it.



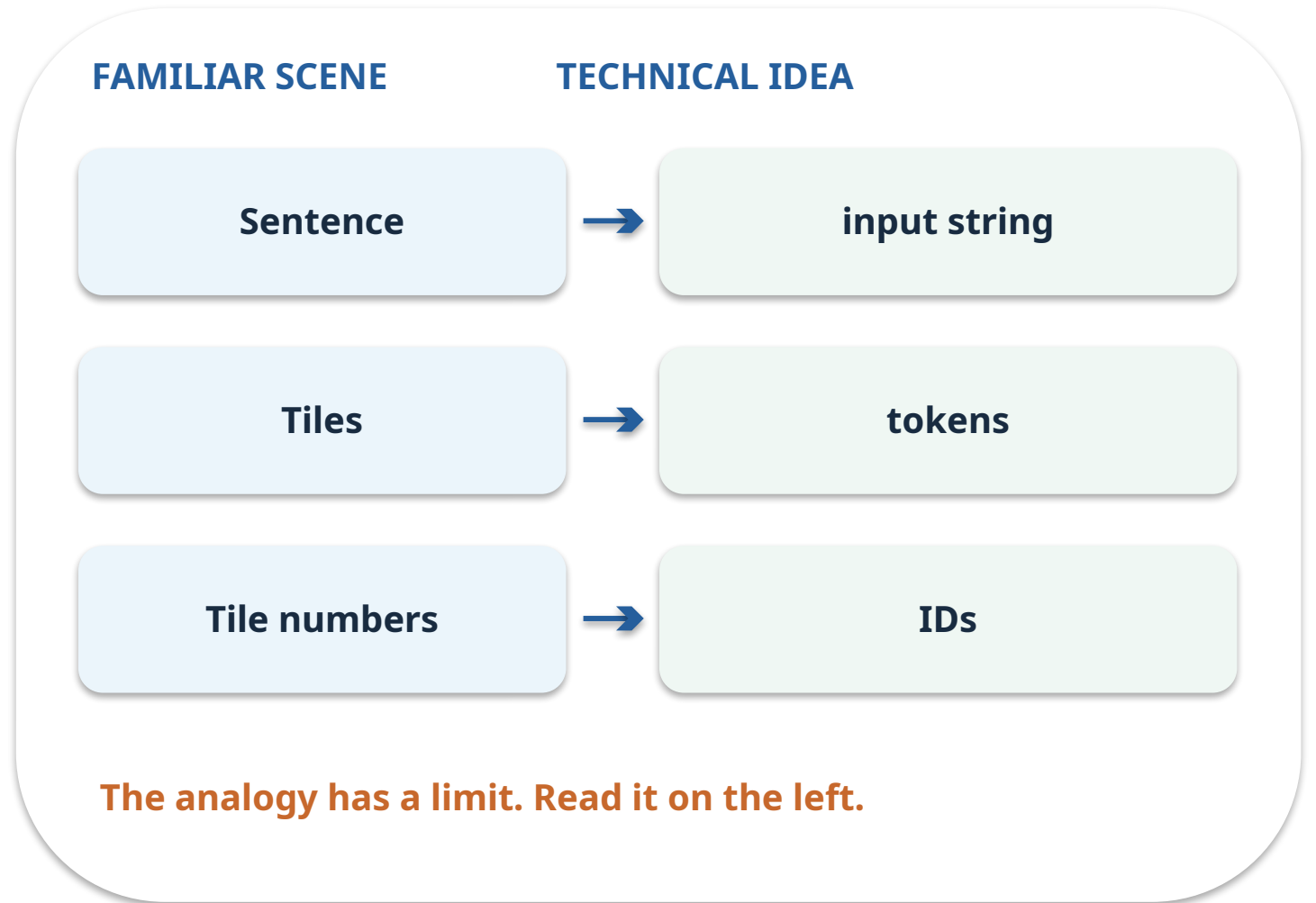
What matches the technical process?

Token · map the parts

MAPPING / WEEK 04

Match each everyday part to the technical part.

Analogy limit: Token pieces need not be complete words; the vocabulary is model-specific.



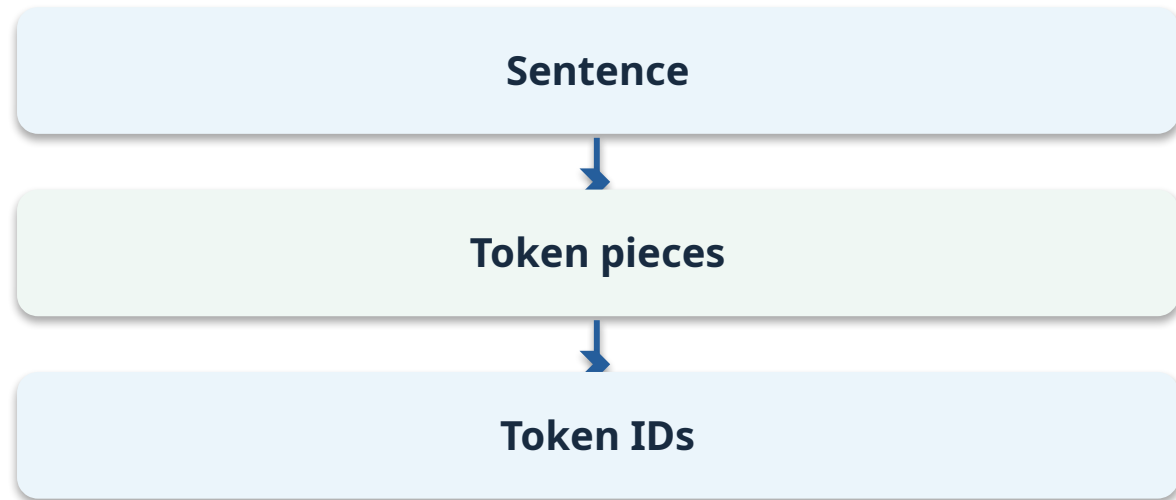
Token • how it works

MECHANISM / WEEK 04

A model processes pieces of text represented as IDs. A word may use more than one token.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Token • work one case

WORKED / WEEK 04

“unbelievable” might be one piece or several; record the tokenizer used before counting.

Point to the evidence before giving your conclusion.

GIVEN

“unbelievable” might be one piece or several; record the tokenizer used before counting.

QUESTION

Why can word count differ from token count?

SUPPORTED ANSWER

A tokenizer may split a word into several pieces.

Token • answer and explain

REVEAL / WEEK 04

A tokenizer may split a word into several pieces.

Tempting mistake: One word always equals one token.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

One word always equals one token.

SUPPORTED EXPLANATION

A tokenizer may split a word into several pieces.

Token pieces need not be complete words; the vocabulary is model-specific.

2. Predict · Attention

QUESTION / WEEK 04

Do weights identify a proven causal explanation?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A No. They show one computation, not the whole model's reason.

B The largest attention cell proves the final answer.

Which source or test could settle this?

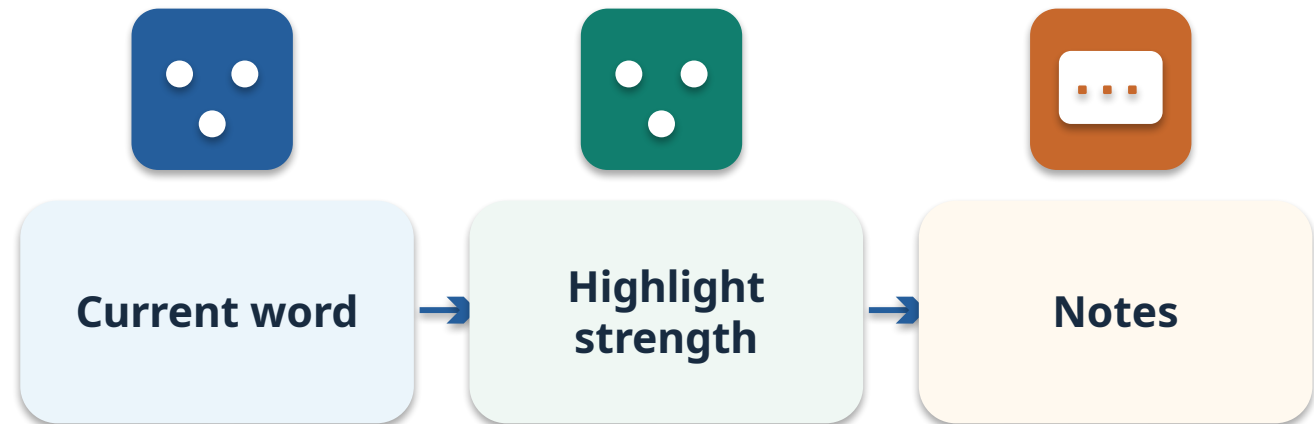
Attention · a familiar picture

ANALOGY / WEEK 04

A reader highlights earlier words that matter to the current word.

What matches our case, and where does the analogy stop?

A reader highlights earlier words that matter to the current word.



What matches the technical process?

Attention · map the parts

MAPPING / WEEK 04

Match each everyday part to the technical part.

Analogy limit: Attention weights alone are not a complete explanation of model decisions.

FAMILIAR SCENE

Current word

Highlight strength

Notes

TECHNICAL IDEA

query

attention weight

weighted context

The analogy has a limit. Read it on the left.

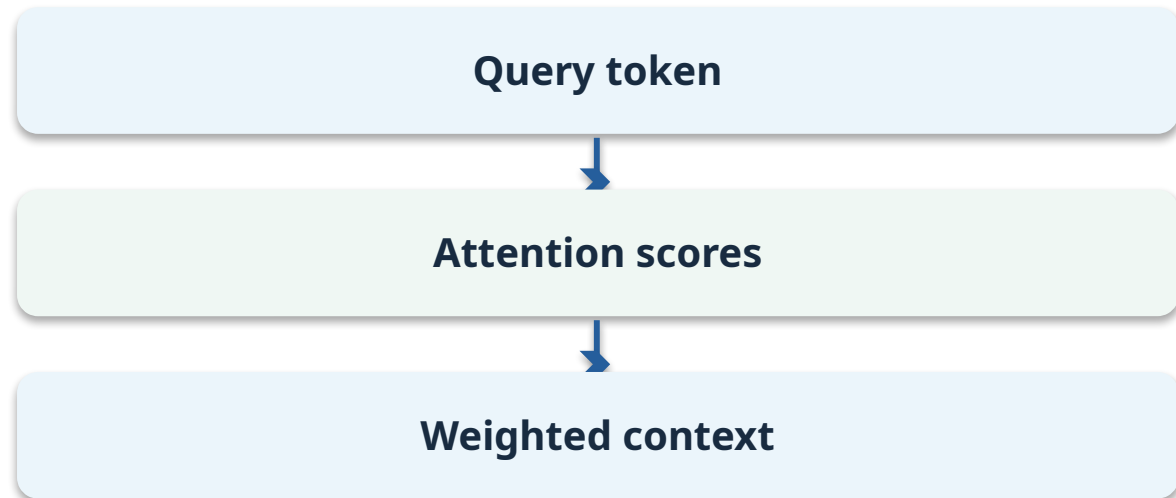
Attention · how it works

MECHANISM / WEEK 04

Each query uses scores to weight available tokens; the weighted combination carries contextual information.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Attention · work one case

WORKED / WEEK 04

Scores [1,2,0] yield about [0.245,0.665,0.090]; the middle token gets most weight.

Point to the evidence before giving your conclusion.

GIVEN

Scores [1,2,0] yield about [0.245,0.665,0.090]; the middle token gets most weight.

QUESTION

Do weights identify a proven causal explanation?

SUPPORTED ANSWER

No. They show one computation, not the whole model's reason.

Attention · answer and explain

REVEAL / WEEK 04

No. They show one computation, not the whole model's reason.

Tempting mistake: The largest attention cell proves the final answer.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The largest attention cell proves the final answer.

SUPPORTED EXPLANATION

No. They show one computation, not the whole model's reason.

Attention weights alone are not a complete explanation of model decisions.

3. Predict · Mask

QUESTION / WEEK 04

What happens to the forbidden position?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Its weight is zero; visible positions are renormalised.

B Masking deletes a token from the vocabulary.

Which source or test could settle this?

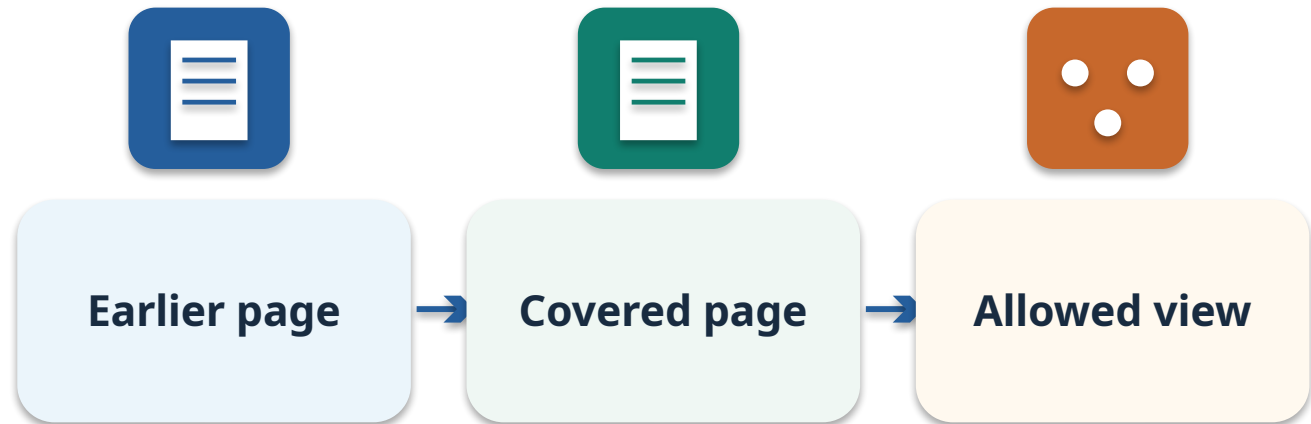
Mask · a familiar picture

ANALOGY / WEEK 04

An exam candidate may consult earlier pages, not tomorrow's answer sheet.

What matches our case, and where does the analogy stop?

An exam candidate may consult earlier pages, not tomorrow's answer sheet.



What matches the technical process?

Mask · map the parts

MAPPING / WEEK 04

Match each everyday part to the technical part.

Analogy limit: A mask is an enforced attention constraint; it does not ensure factual accuracy.

FAMILIAR SCENE

Earlier page

Covered page

Allowed view

TECHNICAL IDEA

past token

masked future token

causal context

The analogy has a limit. Read it on the left.

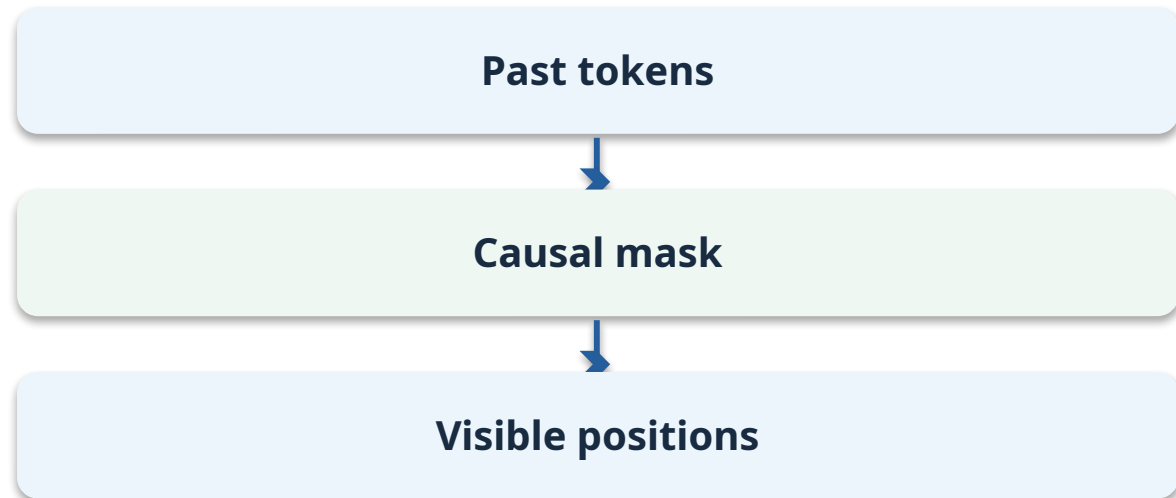
Mask · how it works

MECHANISM / WEEK 04

A decoder mask prevents a token from using future tokens while generating left to right.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Mask · work one case

WORKED / WEEK 04

Mask the third of three scores [1,2,0]; renormalised weights are about [0.269,0.731,0].

Point to the evidence before giving your conclusion.

GIVEN

Mask the third of three scores [1,2,0]; renormalised weights are about [0.269,0.731,0].

QUESTION

What happens to the forbidden position?

SUPPORTED ANSWER

Its weight is zero; visible positions are renormalised.

Mask · answer and explain

REVEAL / WEEK 04

Its weight is zero; visible positions are renormalised.

Tempting mistake: Masking deletes a token from the vocabulary.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Masking deletes a token from the vocabulary.

SUPPORTED EXPLANATION

Its weight is zero; visible positions are renormalised.

A mask is an enforced attention constraint; it does not ensure factual accuracy.

4. Predict · Model roles

QUESTION / WEEK 04

Which role suits translation in this simplified comparison?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A An encoder reads source text and a decoder writes target text.

B An encoder always generates the final long reply.

Which source or test could settle this?

Model roles • a familiar picture

ANALOGY / WEEK 04

A reader, a writer, and a reader-writer have different jobs.

What matches our case, and where does the analogy stop?

A reader, a writer, and a reader-writer have different jobs.



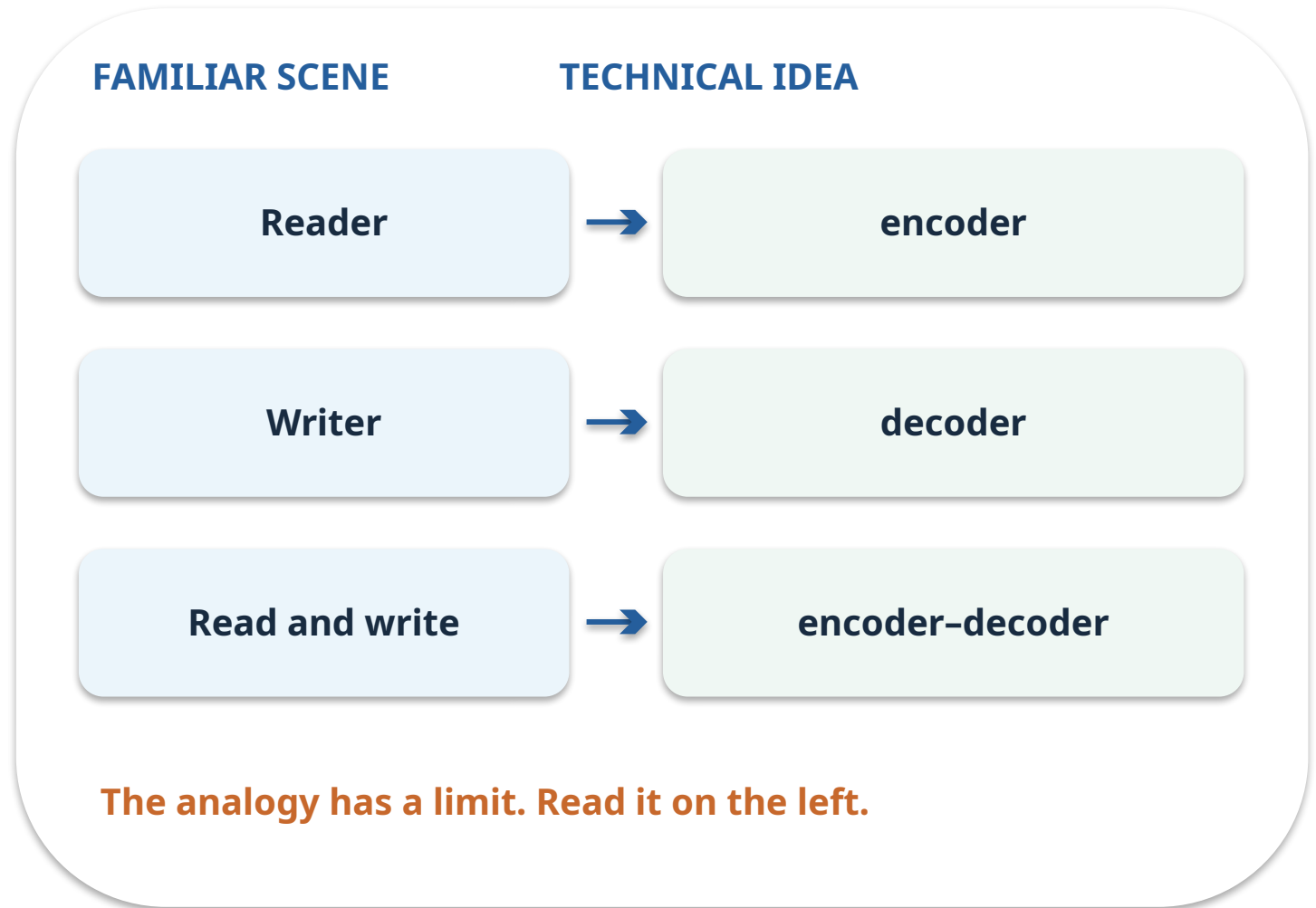
What matches the technical process?

Model roles · map the parts

MAPPING / WEEK 04

Match each everyday part to the technical part.

Analogy limit: Many current systems combine components; this is a useful role sketch, not a full taxonomy.



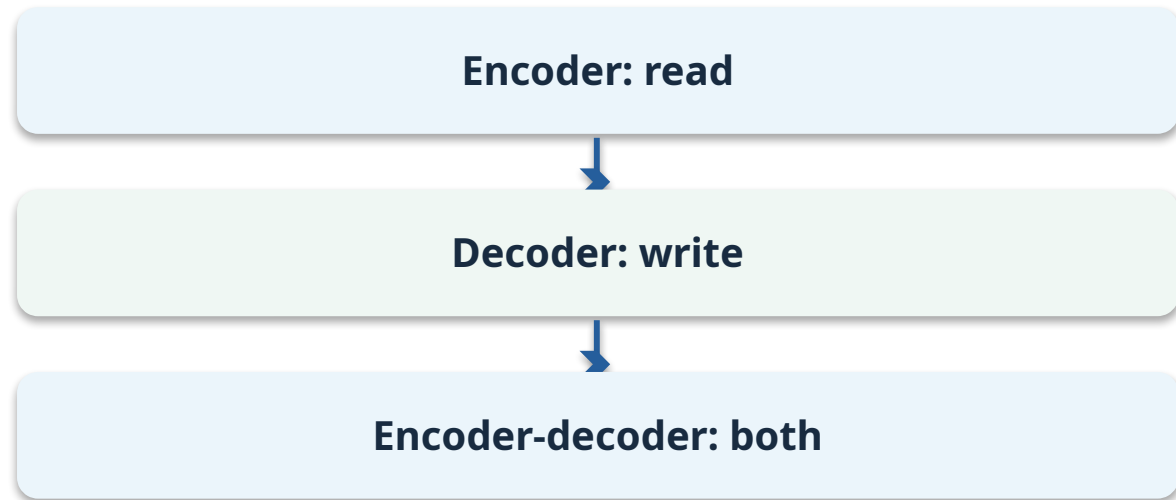
Model roles • how it works

MECHANISM / WEEK 04

Encoders read inputs; decoders generate; encoder-decoder models read then generate from a separate component.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Model roles • work one case

WORKED / WEEK 04

Classify a sentence: encoder.
Continue it: decoder. Translate it:
encoder–decoder.

Point to the evidence before
giving your conclusion.

GIVEN

Classify a sentence: encoder. Continue it: decoder. Translate it: encoder–decoder.

QUESTION

Which role suits translation in this simplified comparison?

SUPPORTED ANSWER

An encoder reads source text and a decoder writes target text.

Model roles · answer and explain

REVEAL / WEEK 04

An encoder reads source text and a decoder writes target text.

Tempting mistake: An encoder always generates the final long reply.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

An encoder always generates the final long reply.

SUPPORTED EXPLANATION

An encoder reads source text and a decoder writes target text.

Many current systems combine components; this is a useful role sketch, not a full taxonomy.

Evidence trail: input → result

WORKED / WEEK 04

Softmax weights approximately 0.245, 0.665, 0.090.

If the third token is future text, its masked weight becomes 0.

GIVEN

Scores for three tokens: 1, 2, 0.

QUESTION

Softmax weights approximately 0.245, 0.665, 0.090.

SUPPORTED ANSWER

If the third token is future text, its masked weight becomes 0.

Notebook: run the prepared example

PRACTICE / WEEK 04

Compute attention for three tokens and explain one mask.

Read the original output before editing.

1

Tokenize a tiny three-word classroom example

2

Compute normalized attention weights from simple scores

3

Apply a mask to hide a future word

4

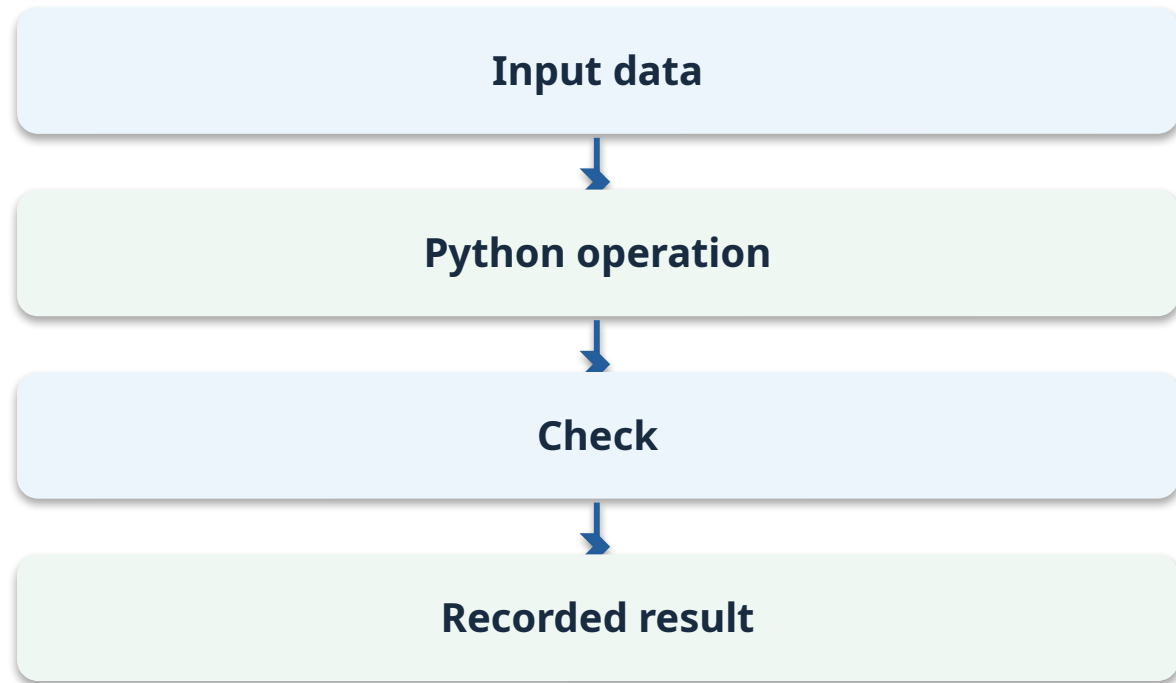
Compare an encoder and a decoder task

Find the input and check in Python

PRACTICE / WEEK 04

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 04

Mask the last token and recompute the weights.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 04

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

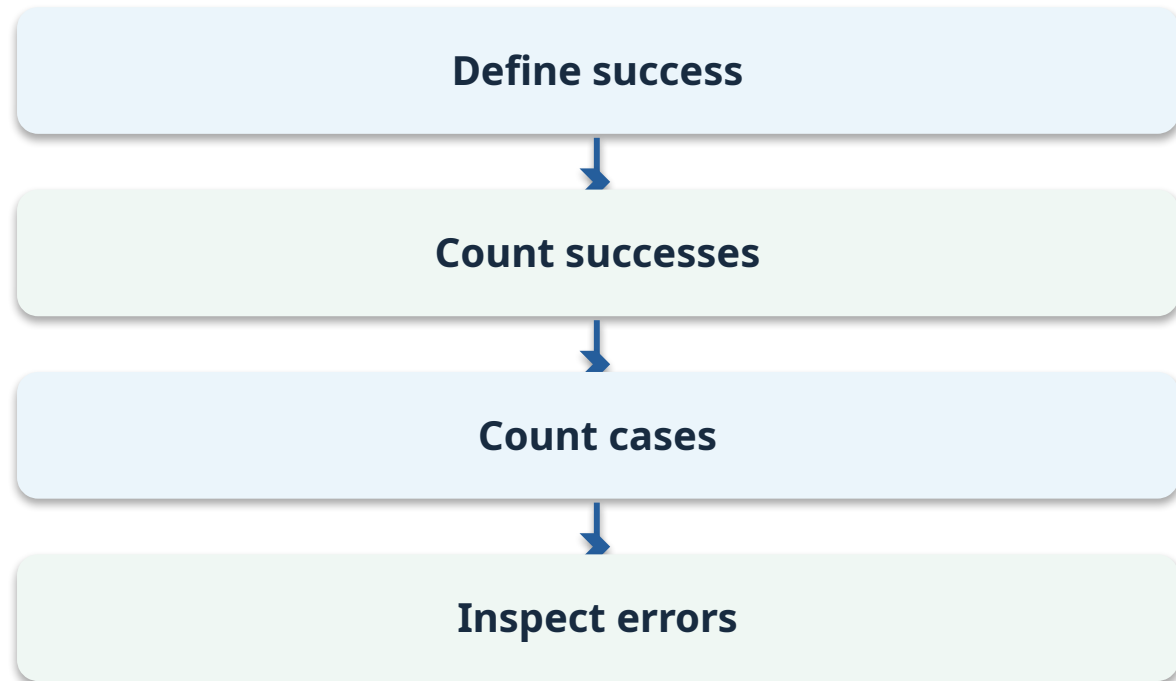
Measure without hiding the denominator

CHECK / WEEK 04

Weights sum to one over visible tokens

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 04

Plausible wrong claim: Attention weights are a complete explanation of a model decision.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Attention weights are a complete explanation of a model decision.

SUPPORTED EXPLANATION

They show one internal weighting operation; they do not prove causal importance or factual correctness.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 04

A drafting decoder has the next word hidden. What happens to that column in its attention grid?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

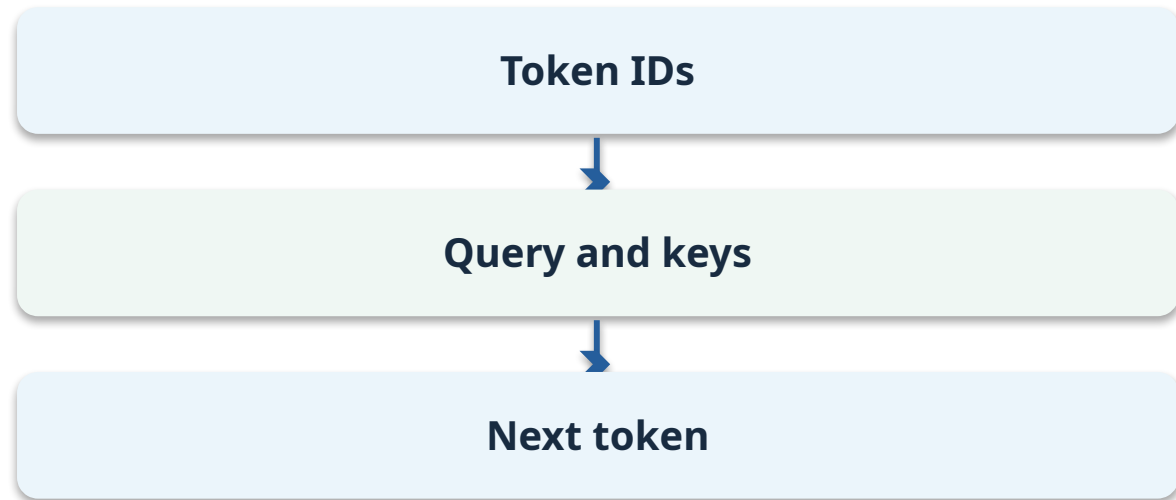
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 04

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 04

Why does a decoder use a causal mask?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 04

For weights 0.25, 0.65, 0.10, which input receives most weight?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

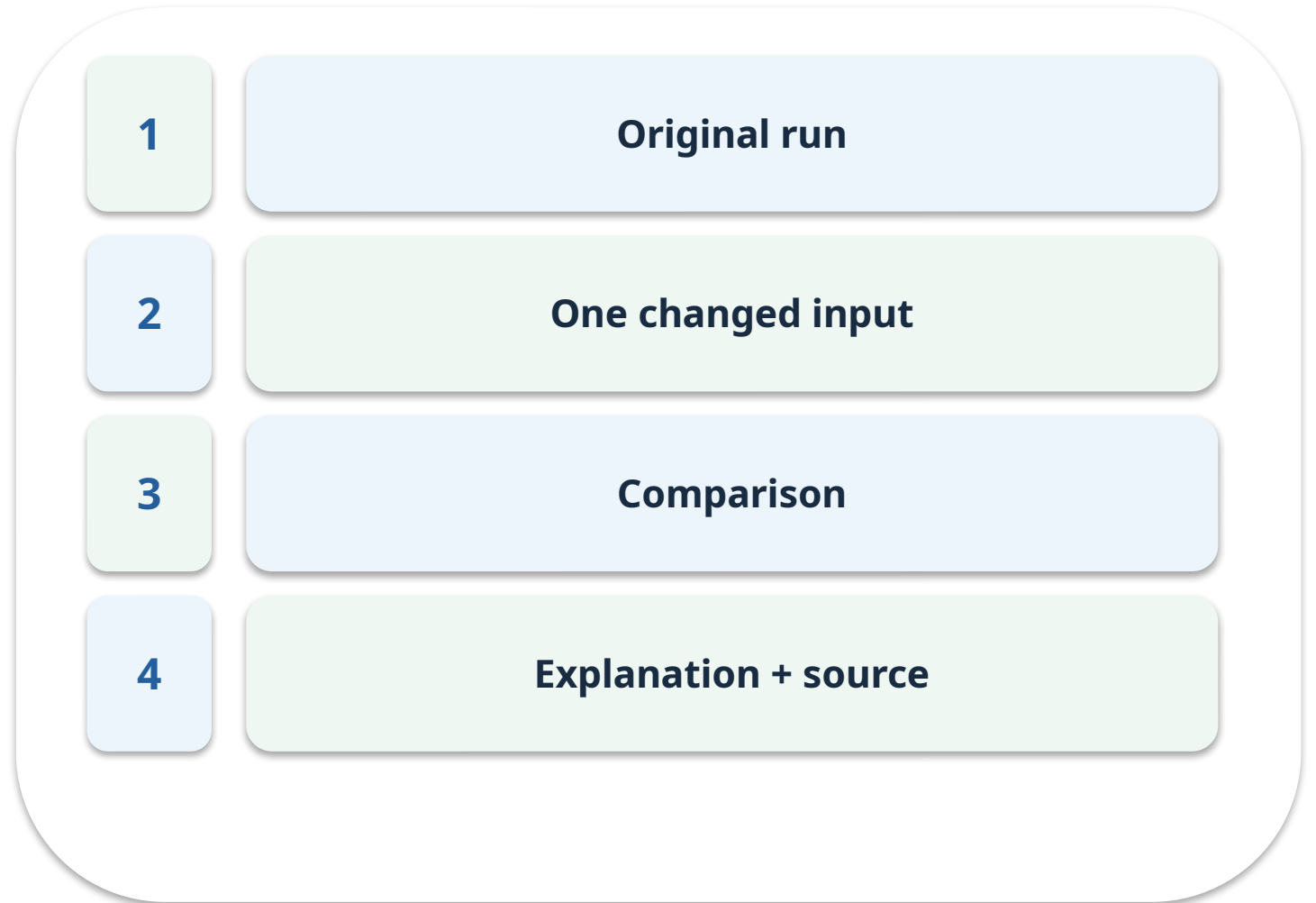
Compare with the Word guide

What to hand in

PRACTICE / WEEK 04

Annotated attention table and one model-role choice.

Include the one changed input and an error explanation.



Three ideas to recall next week

CHECK / WEEK 04

Tokenise, score visible context, and choose the model role for reading or writing.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Token

2

Attention

3

Mask

4

Model roles